

Android - Network Connection

Android lets your application connect to the internet or any other local network and allows you to perform network operations.

A device can have various types of network connections. This topic we focus on using either a Wi-Fi or a mobile network connection.

Checking Network Connection

Before you perform any network operations, you must first check that are you connected to that network or internet etc. For this android provides **ConnectivityManager** class. You need to instantiate an object of this class by calling **getSystemService()** method. Its syntax is given below:

```
ConnectivityManager check = (ConnectivityManager)
this.context.getSystemService(Context.CONNECTIVITY_SERVICE);
```

Once you instantiate the object of **ConnectivityManager** class, you can use **getAllNetworkInfo** method to get the information of all the networks. This method returns an array of **NetworkInfo**. So you have to receive it like this.

```
NetworkInfo[] info = check.getAllNetworkInfo();
```

The last thing you need to do is to check **Connected State** of the network. Its syntax is given below:

```
for (int i = 0; i<info.length; i++){
    if (info[i].getState() == NetworkInfo.State.CONNECTED){
        Toast.makeText(context, "Internet is connected
        Toast.LENGTH_SHORT).show();
    }
}
```

Apart from this connected states, there are other states a network can achieve. They are listed below:

S. No	State
1	Connecting
2	Disconnected
3	Disconnecting
4	Suspended
5	Unknown

Performing Network Operations

After checking that you are connected to the internet, you can perform any network operation. Here we are fetching the html of a website from a url.

Android provides **URLConnection** and **URL** class to handle these operations. You need to instantiate an object of URL class by providing the link of website. Its syntax is as follows –

```
String link = "http://www.google.com";  
URL url = new URL(link);
```

After that you need to call **openConnection** method of url class and receive it in a HttpURLConnection object. After that you need to call the **connect** method of HttpURLConnection class.

```
URLConnection conn = (URLConnection) url.openConnection();  
conn.connect();
```

And the last thing you need to do is to fetch the HTML from the website. For this you will use **InputStream** and **BufferedReader** class. Its syntax is given below –

```
InputStream is = conn.getInputStream();  
BufferedReader reader = new BufferedReader(new InputStreamReader(is, "UTF-8"));  
String webPage = "", data="";  
  
while ((data = reader.readLine()) != null){  
    webPage += data + "\n";  
}
```

Apart from this connect method, there are other methods available in HttpURLConnection class. They are listed below –

Sr.No	Method & description
1	disconnect() This method releases this connection so that its resources may be either reused or closed
2	getRequestMethod() This method returns the request method which will be used to make the request to the remote HTTP server
3	getResponseCode() This method returns response code returned by the remote HTTP server
4	setRequestMethod(String method) This method Sets the request command which will be sent to the remote HTTP server
5	usingProxy() This method returns whether this connection uses a proxy server or not

Source: - https://www.tutorialspoint.com/android/android_network_connection.htm